FASITH AHAMED F

+91 9363573142 | fasith23ahamed@gmail.com | Website | Linkedin | Github

EDUCATION

Indian Institute of Technology Madras, Chennai, India *CGPA* ***8.62/10.00***
Dual Degree in Engineering Design, BTech (Engineering Design) and MTech (Automotive) *2020 – 2025*

EXPERIENCE

Software Engineer (GET) *July 2025 – June 2026*
OLA

- **Home Screen Revamp:** Rebuilt the home screen of the OLA Consumer App in **Swift** with API driven card ordering following **MVVM** architecture & SOLID principle, increasing 20% user transfer to Booking screen
- **Slider Tipping:** Created a custom stepped price slider in **SwiftUI** with geometry-based layout, drag snapping, dynamic segment coloring, haptic feedback, and gesture-based dismissal handling - integrated into UIkit, Coordinators
- **Automated Promotion Generation:** Implemented UTM parameter extraction and deep link routing for universal links, propagating referral source through flows to maintain attribution across user journeys
- Worked with observables and **concurrency** for reactive Home screen card ordering and card updates in Slider Tipping
- Independently built key UI components such as Shimmer animations & released builds via **Jenkins** & **Fastlane**

Dual Degree Thesis *June 2024 – May 2025*
IIT Madras, Guide: Prof. M Ramanathan *AR Tennis Swing Analysis*

- Developed a pipeline for an Augmented Reality setup to project an optimum tennis swing; with **Unity** to display humanoid model, **FastAPI** to broadcast IMU and camera data and **Backend** (**python**) to match 3D pose from a pre-trained model
- Designed a YOLO player/ball detection(0.89/0.73 mAP) on tennis videos to be used in training Pose estimation model

Research Developer Intern *Dec 2023 – May 2024*
Ghost Vision Pvt. Ltd. Documentation

- Developed a sensor fusion solution for an Augmented Reality app to store and retrieve geolocated entities in realtime
- Implemented a **microservices architecture**, includes Broadcast service (**FastAPI, Python**), display (**ReactJS**), data processing (Python) and backend (**.NET, c#, mongoDB**), using **REST** and **Websockets**, to be containerized in **Docker**
- Upsampled GPS data to sync with IMU data using regression over a parametric buffer of 5 yielding 75% accuracy
- Written **Shell Scripts** for the build & installation of ORB-SLAM 2,3 & ROS in a Virtual Machine for geolocation (camera)

Young Research Fellow *Sep 2022 – Aug 2023*
IIT Madras, Guide: Prof. M Ramanathan *3D Reconstruction of Musical Instruments*

- Analyzed musical instrument geometry to relate it to its vibrational properties & frequencies for 3D reconstruction
- Developed a 3D model generator in **Blender** (**Python** scripts) to create Ukulele & Veena; generated a 1500+ dataset
- Selected as one of the **30** students among **300+** candidates for the **Young Research Fellowship 2022** by IIT Madras

PROJECTS

Boids Simulation | Package | Demo ***Swift***, *SwiftUI, Swift Concurrency* | *Aug 2026 - Sep 2026*

- **Spatial Hash** & **Quad Tree** implementation of Boids Simulation, with **Tasks & Actors** for parallelisation & concurrency
- Assistive Menu to change parameters and search algorithms in real-time; Available as an **open-source** Swift Package

Triangulation Visualizer | Website | Source Code *Aug 2023 - Nov 2023*
Course: Algorithms in Computational Geometry *C++, Python, ReactJS, FastAPI, TS*

- Real-time Delaunay Triangulation using convex hulls (**divide & conquer algorithm**) in c++ (DLL processed in python)

Game Dev Hackathons *C#, Unity 3D, Blender*

- Blue Fish Yokohama: Worked on the UI and animations for a puzzle-based game, with pixel art as the USP Steam
- NASA Space Apps: Led the creation of an interactive maze game for kids involving space exploration Link

TECHNICAL SKILLS

Languages Python, C#, C++, Swift, C, JS | **Softwares** Unity, Unreal Engine 4, Blender, OpenCV, OpenGL, Fusion360
Tools Git, Jenkins, Linux, Docker, Shell Scripts | **Frameworks** .NET, FastAPI, SwiftUI, UIKit, NextJS, TailwindCSS, ReactJS